Differential Equations and Slope Fields

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Verifying solutions to differential equations**

Verify that $y = Ce^{3x}$ is a solution to the differential equation $\frac{dy}{dx} = 3y$ for any constant C.

$\frac{dy}{dx} = 3Ce^{3x} = 3(Ce^{3x}) = 3y$. This confirms $y = Ce^{3x}$ satisfies the equation for any C.

2 **[Foundational]** • **Target: Verifying solutions to differential equations**

Verify that $y = x^2 + C$ is a general solution to $\frac{dy}{dx} = 2x$, and find the particular solution satisfying $y(1) = 5$.

$\frac{dy}{dx} = 2x$ ✓. Using $y(1) = 5$: $1 + C = 5 \Rightarrow C = 4$. Particular solution: $y = x^2 + 4$.

3 **[Foundational]** • **Target: Solving separable differential equations**

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 3$.

$y\,dy = x\,dx \Rightarrow \frac{y^2}{2} = \frac{x^2}{2} + C \Rightarrow y^2 = x^2 + 2C$. Using $y(0) = 3$: $9 = 0 + 2C \Rightarrow 2C = 9$. Solution: $y^2 = x^2 + 9$, i.e. $y = \sqrt{x^2 + 9}$ (positive root, since $y(0) = 3 > 0$).

4 **[Foundational]** • **Target: Solving separable differential equations**

Solve $\frac{dy}{dx} = 2xy$ given $y(0) = 1$.

$\frac{1}{y}dy = 2x\,dx \Rightarrow \ln|y| = x^2 + C$. Using $y(0) = 1$: $\ln 1 = 0 + C \Rightarrow C = 0$. So $\ln y = x^2 \Rightarrow y = e^{x^2}$.

5 **[Foundational]** • **Target: Solving separable differential equations**

Solve $\frac{dy}{dx} = \frac{\cos x}{y^2}$ given $y(0) = 2$.

$y^2\,dy = \cos x\,dx \Rightarrow \frac{y^3}{3} = \sin x + C$. Using $y(0) = 2$: $\frac{8}{3} = 0 + C \Rightarrow C = \frac{8}{3}$. So $y^3 = 3\sin x + 8$, i.e. $y = \sqrt[3]{3\sin x + 8}$.

6 **[Foundational]** • **Target: Slope fields**

For the differential equation $\frac{dy}{dx} = x - y$, find the slope at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$, as would be drawn on a slope field.

At $(0, 0)$: slope $= 0 - 0 = 0$. At $(1, 0)$: slope $= 1 - 0 = 1$. At $(1, 1)$: slope $= 1 - 1 = 0$.

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7 [Foundational] • Target: Slope fields

Explain how a slope field for $\frac{dy}{dx} = ky$ (exponential growth/decay) would look different for $k > 0$ versus $k < 0$.

For $k > 0$: slopes are positive above the x-axis and negative below, steepening as $|y|$ increases — curves grow away from $y = 0$ (unstable equilibrium, exponential growth). For $k < 0$: slopes point toward $y = 0$ from both sides, flattening as $|y| \to 0$ — curves decay toward $y = 0$ (stable equilibrium, exponential decay).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Exponential growth and decay models

A bacteria culture grows according to $\frac{dP}{dt} = 0.2P$, with $P(0) = 500$. Find $P(t)$ and the population after 10 hours.

Solution form: $P(t) = P_0 e^{kt} = 500e^{0.2t}$. $P(10) = 500e^2 \approx 500(7.389) \approx 3694$ bacteria.

9 [Intermediate] • Target: Exponential growth and decay models

A radioactive substance decays according to $\frac{dA}{dt} = -0.05A$. If $A(0) = 100$ grams, find how long it takes to decay to 50 grams (the half-life).

$A(t) = 100e^{-0.05t}$. Set $50 = 100e^{-0.05t}$: $0.5 = e^{-0.05t}$, $\ln(0.5) = -0.05t$, $t = \frac{\ln(0.5)}{-0.05} = \frac{-0.693}{-0.05} \approx 13.86$ years.

10 [Intermediate] • Target: Matching slope fields to differential equations

A slope field shows horizontal segments (slope 0) along the entire x-axis and along the entire y-axis, with slopes elsewhere resembling the product xy. Identify the differential equation this slope field represents.

$\frac{dy}{dx} = xy$ — slope is 0 whenever $x = 0$ or $y = 0$, matching the described field.

11 [Intermediate] • Target: Newton's Law of Cooling

An object cools according to $\frac{dT}{dt} = -k(T - 70)$, where $70°\text{F}$ is room temperature. If $T(0) = 200°\text{F}$ and $k = 0.1$, find $T(t)$.

Let $y = T - 70$: $\frac{dy}{dt} = -0.1y \Rightarrow y = y_0 e^{-0.1t}$. $y_0 = 200 - 70 = 130$. So $T(t) = 70 + 130e^{-0.1t}$.

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12 [Intermediate] • Target: Visualizing an exponential growth solution

The graph shows the bacteria population $P(t) = 500e^{0.2t}$ from the earlier problem. Use the graph to estimate how long it takes for the population to double from its initial value of 500, then verify algebraically using $\ln 2$.

The graph crosses $P = 1000$ at approximately $t \approx 3.5$. Algebraically: $1000 = 500e^{0.2t}$ gives $2 = e^{0.2t}$, so $t = \frac{\ln 2}{0.2} \approx 3.47$, matching the graphical estimate.

13 [Intermediate] • Target: A logistic-style differential equation

A population satisfies $\frac{dP}{dt} = 0.1P\left(1 - \frac{P}{200}\right)$. Find the equilibrium solutions (where $\frac{dP}{dt} = 0$), and explain which one is stable for a population starting near $P = 10$.

Setting $\frac{dP}{dt} = 0$: $P = 0$ or $P = 200$. Starting near $P = 10$ (close to 0 but positive), the population will grow toward the stable equilibrium $P = 200$ (the carrying capacity), since $P = 0$ is unstable for any positive starting population.

14 [Intermediate] • Target: Additional separable equations

Solve $\frac{dy}{dx} = \frac{y}{x}$ given $y(1) = 6$.

$\frac{1}{y}dy = \frac{1}{x}dx \Rightarrow \ln|y| = \ln|x| + C \Rightarrow y = Ax$ for a constant A. Using $y(1) = 6$: $A = 6$: $y = 6x$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Additional separable equations • Cross-Lesson: 8.01 Area Between Curves

Solve $\frac{dy}{dx} = \frac{1}{y}$ given $y(0) = 4$.

$y\,dy = dx \Rightarrow \frac{y^2}{2} = x + C$. Using $y(0) = 4$: $8 = 0 + C \Rightarrow C = 8$. So $y^2 = 2x + 16$, i.e. $y = \sqrt{2x + 16}$ (positive root, since $y(0) = 4 > 0$).

16 [Challenging] • Target: A mixing problem modeled by a differential equation • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

A tank contains 50 L of pure water. Brine with concentration 3 kg/L flows in at 2 L/min, and the well-mixed solution drains at the same rate. If $y(t)$ is the amount of salt (kg) at time t, the differential equation is $\frac{dy}{dt} = 6 - \frac{2y}{50} = 6 - \frac{y}{25}$. Find the equilibrium value of y, and explain what it represents physically.

Setting $\frac{dy}{dt} = 0$: $6 = \frac{y}{25}$ gives $y = 150$ kg. This represents the long-run salt content: eventually the rate of salt entering (6 kg/min from the inflow) balances the rate leaving, keeping the amount constant at 150 kg.

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17 [Challenging] • Target: Exponential growth and decay models • Cross-Lesson: 1.01 Limits Graphically and Numerically

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 20$. Find $y(15)$ (the positive root).

Final Answer: 25. $y\,dy = x\,dx \Rightarrow y^2 = x^2 + 2C$. Using $y(0) = 20$: $2C = 400$. So $y^2 = x^2 + 400$; $y(15) = \sqrt{(15)^2 + 400} = \sqrt{625} = 25$.

18 [Challenging] • Target: Matching slope fields to differential equations • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 12$. Find $y(-5)$ (the positive root).

Final Answer: 13. $y\,dy = x\,dx \Rightarrow y^2 = x^2 + 2C$. Using $y(0) = 12$: $2C = 144$. So $y^2 = x^2 + 144$; $y(-5) = \sqrt{(-5)^2 + 144} = \sqrt{169} = 13$.

19 [Challenging] • Target: Newton's Law of Cooling • Cross-Lesson: 2.01 The Definition of the Derivative

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 4$. Find $y(3)$ (the positive root).

Final Answer: 5. $y\,dy = x\,dx \Rightarrow y^2 = x^2 + 2C$. Using $y(0) = 4$: $2C = 16$. So $y^2 = x^2 + 16$; $y(3) = \sqrt{(3)^2 + 16} = \sqrt{25} = 5$.

20 [Challenging] • Target: Visualizing an exponential growth solution • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 24$. Find $y(-10)$ (the positive root).

Final Answer: 26. $y\,dy = x\,dx \Rightarrow y^2 = x^2 + 2C$. Using $y(0) = 24$: $2C = 576$. So $y^2 = x^2 + 576$; $y(-10) = \sqrt{(-10)^2 + 576} = \sqrt{676} = 26$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Verifying solutions to differential equations

For the differential equation $\frac{dy}{dx} = -3x + 5y$, find the slope at the point $(-1, -4)$, as would be drawn on a slope field.

a. -19

b. -17 ✓

c. -15

d. 23

Slope $= -3(-1) - 20 = 3 - 20 = -17$.

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22 [Foundational] • Target: Solving separable differential equations

For the differential equation $\frac{dy}{dx} = -5x$, find the slope at the point $(0, -3)$, as would be drawn on a slope field.

a. -2

b. 1

c. 0 ✓

d. 2

Slope $= -5(0) = 0 = 0$.

23 [Foundational] • Target: Slope fields

For the differential equation $\frac{dy}{dx} = -x - 3y$, find the slope at the point $(3, 3)$, as would be drawn on a slope field.

a. 6

b. -14

c. -10

d. -12 ✓

Slope $= -1(3) - 9 = -3 - 9 = -12$.

24 [Foundational] • Target: Exponential growth and decay models

For the differential equation $\frac{dy}{dx} = -3x + 5y$, find the slope at the point $(3, 1)$, as would be drawn on a slope field.

a. -4 ✓

b. -14

c. -6

d. -2

Slope $= -3(3) + 5 = -9 + 5 = -4$.

25 [Foundational] • Target: Matching slope fields to differential equations

For the differential equation $\frac{dy}{dx} = -5x + y$, find the slope at the point $(-2, -3)$, as would be drawn on a slope field.

a. 13

b. 7 ✓

c. 9

d. 5

Slope $= -5(-2) - 3 = 10 - 3 = 7$.

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26 [Foundational] • Target: Newton's Law of Cooling

For the differential equation $\frac{dy}{dx} = 0x + 2y$, find the slope at the point $(4, -4)$, as would be drawn on a slope field.

a. -10

b. 8

c. -8 ✓

d. -6

Slope $= 0(4) - 8 = 0 - 8 = -8$.

27 [Foundational] • Target: Visualizing an exponential growth solution

For the differential equation $\frac{dy}{dx} = -3x + 2y$, find the slope at the point $(-4, 4)$, as would be drawn on a slope field.

a. 20 ✓

b. 18

c. 4

d. 22

Slope $= -3(-4) + 8 = 12 + 8 = 20$.

28 [Foundational] • Target: A logistic-style differential equation

For the differential equation $\frac{dy}{dx} = -x + 2y$, find the slope at the point $(-4, -2)$, as would be drawn on a slope field.

a. 8

b. 0 ✓

c. -2

d. 2

Slope $= -1(-4) - 4 = 4 - 4 = 0$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Additional separable equations

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 12$. Find $y(-9)$ (the positive root).

a. 12

b. 9

c. 14

d. 15 ✓

$y\,dy = x\,dx \Rightarrow y^2 = x^2 + 2C$. Using $y(0) = 12$: $2C = 144$. So $y^2 = x^2 + 144$;
$y(-9) = \sqrt{(-9)^2 + 144} = \sqrt{225} = 15$.

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30 [Intermediate] • Target: A mixing problem modeled by a differential equation

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 40$. Find $y(-9)$ (the positive root).

a. 40

b. 39

c. 41 ✓

d. 9

$y\,dy = x\,dx \Rightarrow y^2 = x^2 + 2C$. Using $y(0) = 40$: $2C = 1600$. So $y^2 = x^2 + 1600$;
$y(-9) = \sqrt{(-9)^2 + 1600} = \sqrt{1681} = 41$.

31 [Intermediate] • Target: Slope Fields & Solution Curves

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 12$. Find $y(9)$ (the positive root).

a. 9

b. 12

c. 15 ✓

d. 14

$y\,dy = x\,dx \Rightarrow y^2 = x^2 + 2C$. Using $y(0) = 12$: $2C = 144$. So $y^2 = x^2 + 144$;
$y(9) = \sqrt{(9)^2 + 144} = \sqrt{225} = 15$.

32 [Intermediate] • Target: Separation of Variables

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 7$. Find $y(-24)$ (the positive root).

a. 25 ✓

b. 28

c. 7

d. 24

$y\,dy = x\,dx \Rightarrow y^2 = x^2 + 2C$. Using $y(0) = 7$: $2C = 49$. So $y^2 = x^2 + 49$;
$y(-24) = \sqrt{(-24)^2 + 49} = \sqrt{625} = 25$.

33 [Intermediate] • Target: Exponential Growth & Decay Models

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 16$. Find $y(12)$ (the positive root).

a. 12

b. 19

c. 16

d. 20 ✓

$y\,dy = x\,dx \Rightarrow y^2 = x^2 + 2C$. Using $y(0) = 16$: $2C = 256$. So $y^2 = x^2 + 256$;
$y(12) = \sqrt{(12)^2 + 256} = \sqrt{400} = 20$.

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34 [Intermediate] • Target: Verifying solutions to differential equations

A quantity grows according to $\frac{dP}{dt} = \frac{3}{20}P$, with $P(0) = 425$. Find $P(12)$, rounded to the nearest whole number.

a. 2957

b. 2571 ✓

c. 2185

d. 2213

$P(t) = P_0 e^{kt} = 425e^{\frac{3}{20}t}$. $P(12) = 425e^{\frac{3}{20}(12)} \approx 2571$.

35 [Intermediate] • Target: Solving separable differential equations

A quantity grows according to $\frac{dP}{dt} = \frac{1}{4}P$, with $P(0) = 805$. Find $P(10)$, rounded to the nearest whole number.

a. 9807 ✓

b. 8336

c. 7638

d. 11278

$P(t) = P_0 e^{kt} = 805e^{\frac{1}{4}t}$. $P(10) = 805e^{\frac{1}{4}(10)} \approx 9807$.

36 [Intermediate] • Target: Slope fields

A substance decays according to $\frac{dA}{dt} = -\frac{1}{10}A$. If $A(0) = 126$ grams, find how long it takes to decay to 84 grams, to one decimal place.

a. 9.1

b. 2.8

c. 4.1 ✓

d. 5.3

$A(t) = 126e^{-\frac{1}{10}t}$. Set $84 = 126e^{-\frac{1}{10}t}$: $t = \frac{\ln(\frac{2}{3})}{-\frac{1}{10}} \approx 4.1$.

37 [Intermediate] • Target: Exponential growth and decay models

A substance decays according to $\frac{dA}{dt} = -\frac{1}{20}A$. If $A(0) = 267$ grams, find how long it takes to decay to 89 grams, to one decimal place.

a. 22.0 ✓

b. 28.6

c. 27.0

d. 15.4

$A(t) = 267e^{-\frac{1}{20}t}$. Set $89 = 267e^{-\frac{1}{20}t}$: $t = \frac{\ln(\frac{1}{3})}{-\frac{1}{20}} \approx 22.0$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Matching slope fields to differential equations** • **Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem**

A tank's salt content $y(t)$ (kg) satisfies $\frac{dy}{dt} = 9 - \frac{y}{25}$. Find the equilibrium value of y.

a. 216

b. 34

c. 225 ✓

d. 250

Setting $\frac{dy}{dt} = 0$: $9 = \frac{y}{25}$ gives $y = 9(25) = 225$ kg.

39 **[Challenging]** • **Target: Newton's Law of Cooling** • **Cross-Lesson: 2.01 The Definition of the Derivative**

A tank's salt content $y(t)$ (kg) satisfies $\frac{dy}{dt} = 10 - \frac{y}{30}$. Find the equilibrium value of y.

a. 300 ✓

b. 290

c. 330

d. 40

Setting $\frac{dy}{dt} = 0$: $10 = \frac{y}{30}$ gives $y = 10(30) = 300$ kg.

40 **[Challenging]** • **Target: Visualizing an exponential growth solution** • **Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives**

A tank's salt content $y(t)$ (kg) satisfies $\frac{dy}{dt} = 5 - \frac{y}{57}$. Find the equilibrium value of y.

a. 280

b. 62

c. 285 ✓

d. 342

Setting $\frac{dy}{dt} = 0$: $5 = \frac{y}{57}$ gives $y = 5(57) = 285$ kg.

41 **[Challenging]** • **Target: Slope Fields & Solution Curves** • **Cross-Lesson: 3.01 The Chain Rule**

Along which line in the slope field for $\frac{dy}{dx} = 2x + y$ are all slope segments horizontal?

a. $y = -2x$ ✓

b. $y = 2x$

c. $x = 0$

d. $y = 0$

Horizontal slopes have $\frac{dy}{dx} = 0 \Rightarrow 2x + y = 0 \Rightarrow y = -2x$.

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42 [Challenging] • Target: Separation of Variables • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Solve the differential equation $\frac{dy}{dx} = 3x^2y$ with initial condition $y(0) = 5$.

a. $y = 5e^{x^3}$ ✓

b. $y = e^{x^3} + 4$

c. $y = 5e^{3x^2}$

d. $y = x^3 + 5$

Separate variables: $\frac{1}{y}dy = 3x^2dx \Rightarrow \ln|y| = x^3 + C \Rightarrow y = Ae^{x^3}$. Since $y(0) = 5$, $A = 5$, giving $y = 5e^{x^3}$.

43 [Challenging] • Target: Separation of Variables • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

What is the domain of the particular solution $y = \frac{1}{2-x}$ with $y(0) = \frac{1}{2}$?

a. $(-\infty, 2)$ ✓

b. $(2, \infty)$

c. $x \neq 2$

d. All real numbers

The solution must be on an open interval containing the initial point $x_0 = 0$. Since the discontinuity is at $x = 2$, the domain is $(-\infty, 2)$.

44 [Challenging] • Target: Exponential Growth & Decay Models • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

A population grows according to $\frac{dP}{dt} = 0.05P$. If initial population is $1,000$, what is the population at $t = 20$?

a. $1000e$ ✓

b. $1000e^2$

c. 2000

d. $1000(1.05)^{20}$

$P(t) = 1000e^{0.05t}$. For $t = 20$: $P(20) = 1000e^{0.05(20)} = 1000e^1 = 1000e$.

45 [Challenging] • Target: Newton's Law of Cooling • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Newton's Law of Cooling states that $\frac{dT}{dt} = -k(T - T_m)$. What does T_m represent?

a. **The ambient temperature of the surrounding medium** ✓

b. The initial temperature of the object

c. The maximum temperature reached

d. The rate of heat transfer

In Newton's Law of Cooling, T_m is the constant ambient temperature of the surrounding environment.

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